LAB EXPERIMENT RECORD

SET 14: Survey Results

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	P.MOHITH KUMAR REDDY
Register Number	192524223
Date	08-October 2026
Day	Thursday

Aim

To visualize the geographic distribution of cities using population and temperature information.

Dataset

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

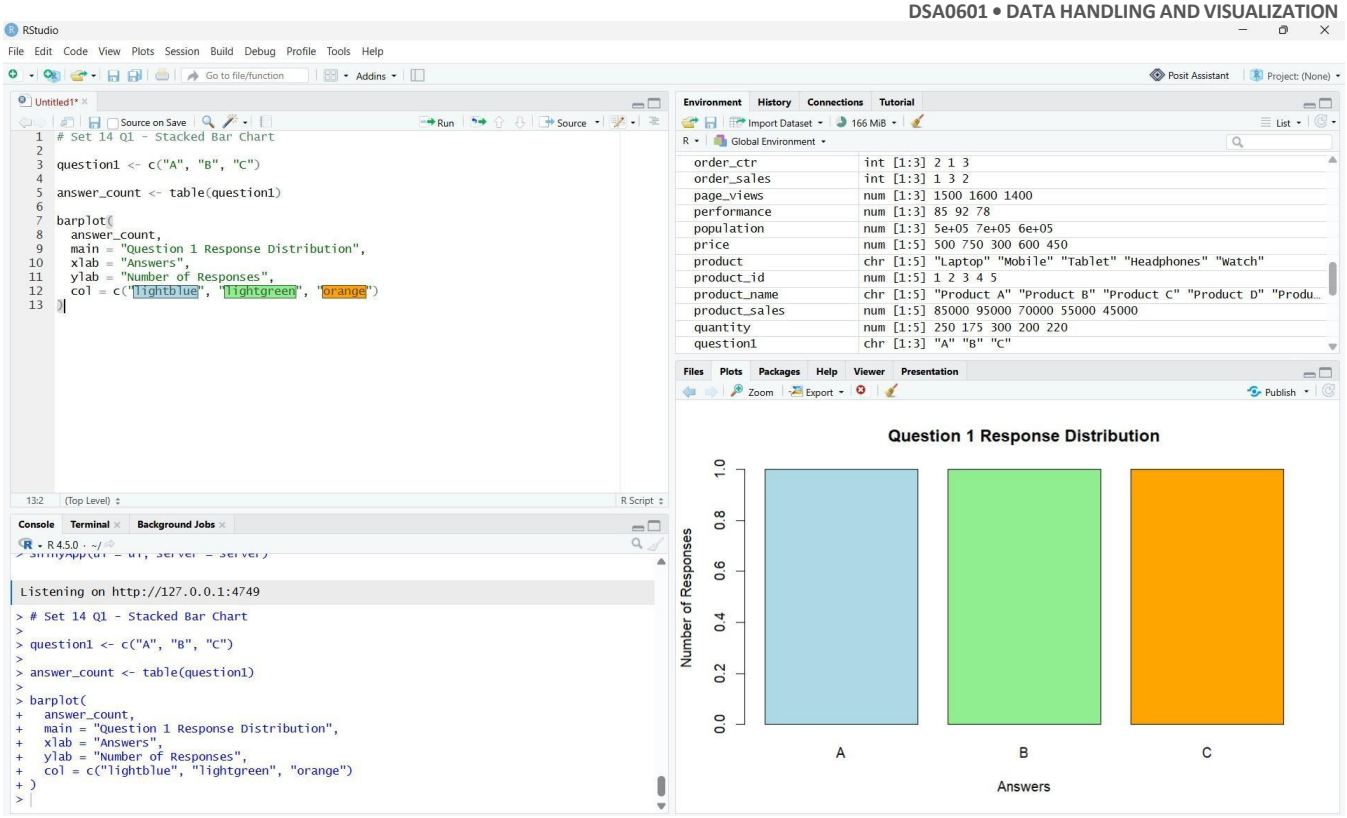
PROGRAM 1: Stacked Bar Chart

To visualize the distribution of answers for Question 1 using a stacked bar chart.

```
# # # Set 14 Q1 - Stacked Bar Chart
question1 <- c("A", "B", "C")


answer_count <- table(question1)


barplot(
  answer_count,
  main = "Question 1 Response Distribution",
  xlab = "Answers",
  ylab = "Number of Responses",
  col = c("lightblue", "lightgreen", "orange")
)
```



Observation:

- **A = 1 response**
- **B = 1 response**
- **C = 1 response**

Each answer has an equal frequency.

PROGRAM 2: Radar Chart

Base R does not have a built-in radar chart, so we can create a **simple radar-style chart** using Base R.

First calculate the total number of A, B, C and D responses for all questions.

```
# Set 14 Q2 - Radar Chart

question1 <- c("A", "B", "C")

question2 <- c("B", "A", "A")

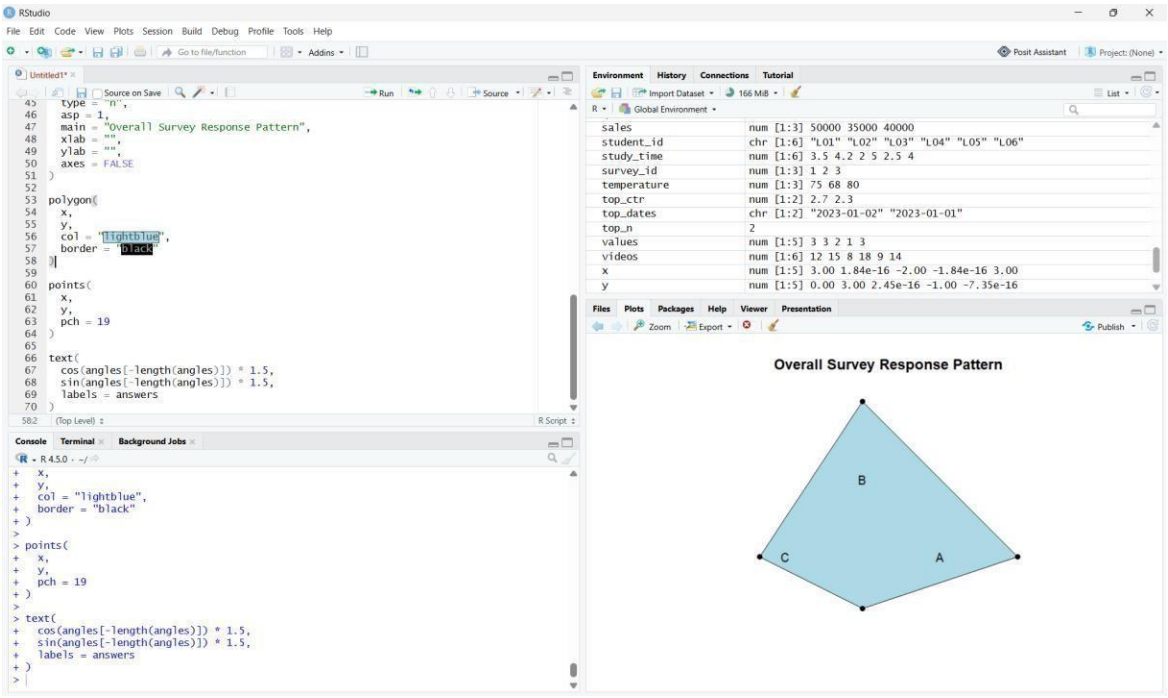
question3 <- c("C", "D", "B")

answers <- c("A", "B", "C", "D")

q1 <- table(factor(question1, levels = answers))

q2 <- table(factor(question2, levels = answers))
```

```
q3 <- table(factor(question3, levels = answers))
overall <- q1 + q2 + q3
values <- as.numeric(overall)
angles <- seq(
  0,
  2 * pi,
  length.out = length(values) + 1
)
values <- c(values, values[1])
x <- values * cos(angles)
y <- values * sin(angles)
plot(
  x,
  y,
  type = "n",
  asp = 1,
  main = "Overall Survey Response Pattern",
  xlab = "",
  ylab = "",
  axes = FALSE
)
polygon(
  x,
  y,
  col = "lightblue",
  border = "black"
)
points(
  x,
  y,
  pch = 19
)text(
  cos(angles[-length(angles)]) * 1.5,
  sin(angles[-length(angles)]) * 1.5,
  labels = answers
```



Observation:

The radar-style chart shows the overall response pattern across A, B, C and D..

PROGRAM 3: Survey Response Table

To display the survey response data for each respondent.

```
# # Set 14 Q3 - Survey Response Table

respondent <- c(1, 2, 3)

question1 <- c("A", "B", "C")

question2 <- c("B", "A", "A")

question3 <- c("C", "D", "B")

survey_table <- data.frame(

  Respondent = respondent,

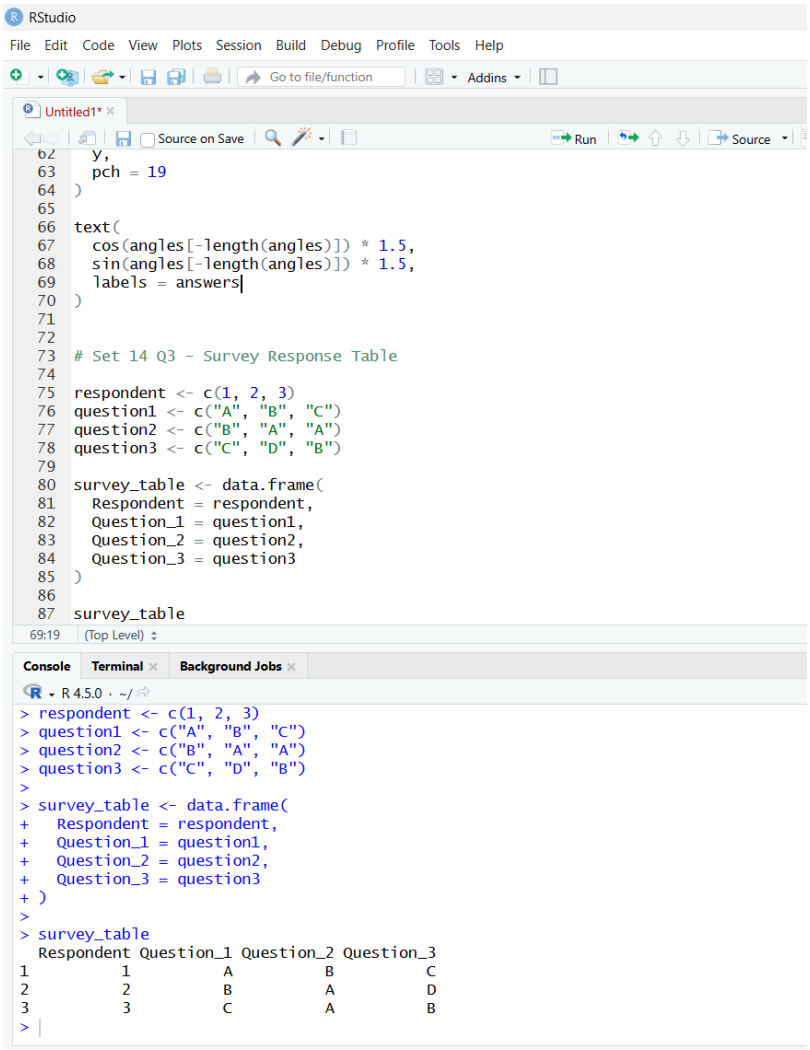
  Question_1 = question1,

  Question_2 = question2,

  Question_3 = question3

)

survey_table
```



Observation

The survey responses of all respondents were successfully displayed in tabular form

PROGRAM 4: Survey Dashboard

```
    library(shiny)


question1 <- c("A", "B", "C")
question2 <- c("B", "A", "A")
question3 <- c("C", "D", "B")



    plot(
      x,
      y,
      type = "n",
      asp = 1,
      main = "Overall Survey Response Pattern",
      axes = FALSE
    )

    polygon(
      x,
      y,
      col = "lightblue"
    )

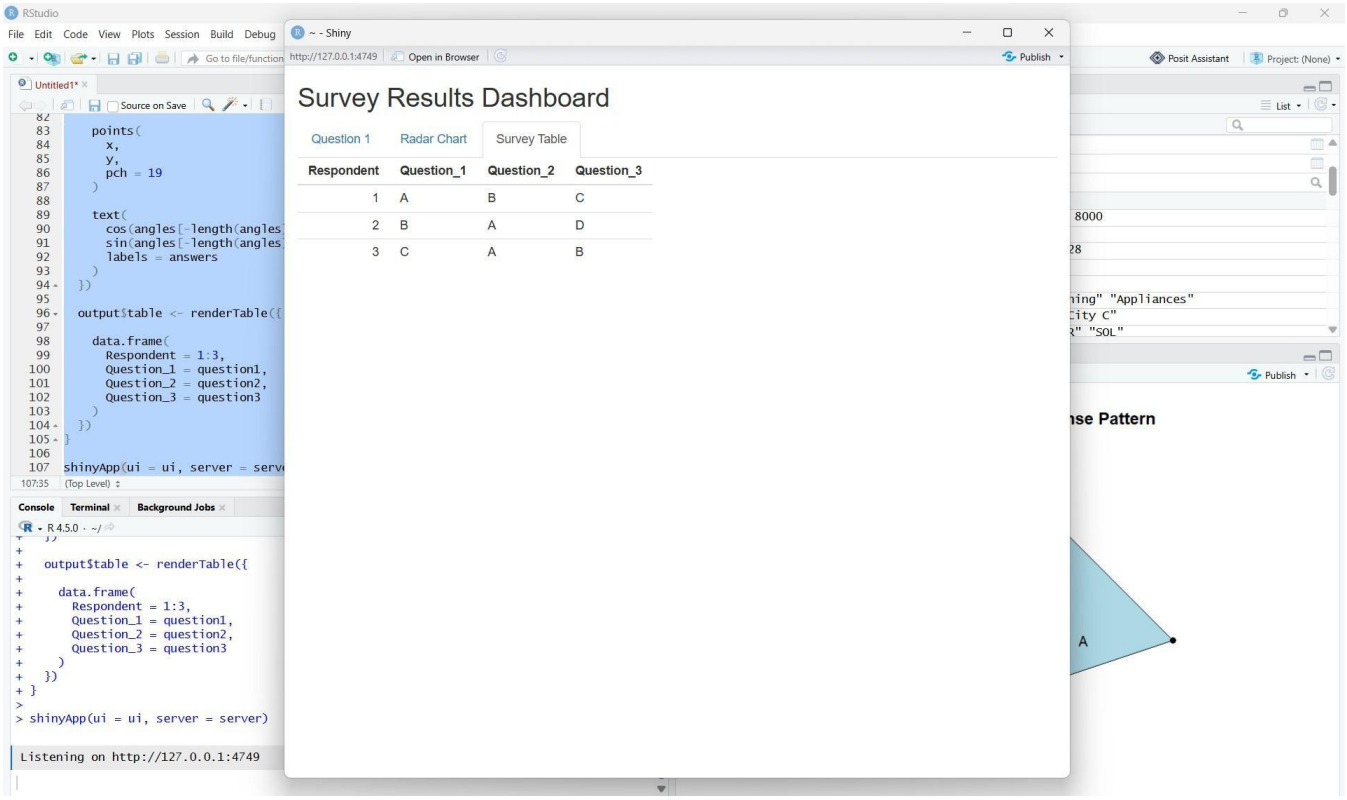
    points(
       x,
      y,
      pch = 19
    )

    text(
      cos(angles[-length(angles)]) * 1.5,
      sin(angles[-length(angles)]) * 1.5,
      labels = answers
    )
  })
```

```
  output$table <- renderTable({


    data.frame(
      Respondent = 1:3,
      Question_1 = question1,
      Question_2 = question2,
      Question_3 = question3
    )
  })
}


shinyApp(ui = ui, server = server)shinyApp(ui = ui, server = server))
```



Output 4: Interactive Monthly Sales Dashboard

Observation: An interactive geographic dashboard was successfully developed in RStudio using Shiny.

.

Result:

The product category sales data was successfully analyzed using a funnel-style chart, pie chart, and table. An An interactive survey dashboard combining the stacked bar chart, radar-style chart and survey table was successfully developed using RStudio Shiny.